



Division of Information Technology & Sciences
Department of Computer & Digital Forensics
FOR-380 – Anti-Forensics
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FOR-380: Final Project Report

Fileless Malware Case Study: Kovter

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Tools:

Below are the tools which were used for the acquisition and forensic analysis of the malware formally referred to as Kovter:

<i>Tool Name</i>	<i>Version</i>	<i>Locality/License</i>	<i>Purpose</i>
MalwareBazaar	–	Website	Used for acquisition of the malware sample
Process hacker 2	2.39 (r124)	GNU GPL, v3	Shows VM Performance, Network Connections, DLL's, Handles, and Activity. Used for understanding the malware's function
PowerShell	5.1.19041.3803 [Desktop]	Microsoft	Used to disable windows defender for the purpose of malware analysis

Sample Information:

The Kovter malware collected for the purpose of this project was collected from MalwareBazaar and has the following information:

File Information	Content:
SHA256 hash:	67012c9555804fbfe82f17ee6d7d09196ed356053e2e778f49a998a4b718d6a6
SHA3-384 hash:	dfef8f339878b56aa0b05f5b732c11e266622ae1827fa5891247ab05e17b798e15e52ee3071201e4d45a84b59ed1686d
SHA1 hash:	40402292ae6494feddea424f0c6e790f6213135d
MD5 hash:	650386014950c28af230457e4686bfe4
humanhash:	wyoming-emma-apart-alaska

File name:	650386014950c28af230457e4686bfe4.exe
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Technique Description:

Kovter, a known clickjacking malware which has been around for a long time, is known majoritively for its fileless architecture. Clickjacking as defined by OWASP is a technique in which “uses multiple transparent or opaque layers to trick a user into clicking on a button or link on another page when they were intending to click on the top level page.” [\[1\]](#). In regards to the MITRE framework its techniques can be classified under the following matrices ‘T1218.010’, ‘T1547.001’, ‘T1055’, and ‘T1059.001’

- **T1218:010**, More commonly referred to as “System Binary Proxy Execution: Regsvr32,” is where a piece of software will use Regscr32.exe in a malicious way in order to allow itself to run for normal operations. This is important as it can be used to bypass application controls. [\[2\]](#)
- **T1547.001**, More commonly known as “Autostart Execution,” is where a malicious piece of software creates registry keys for persistence. This means upon the computer being rebooted the software will restart. This is done to increase the profitability or damage caused by the malware samples. [\[3\]](#)
- **T1055**, Often referred to as “Process Injection,” is a technique in which adversaries will inject code into running and or well known processes in order to allow for execution. This often allows for the malware and or software to gain elevated privileges and evade security measures. [\[4\]](#)
- **T1059.001** is when processes may use powershell scripts for execution of their malware. Given it is a well known program and is installed natively on most windows devices it is a logical attack surface. One major point of note is that this will often be targeted through Assembly code, using the .NET framework, rather than being invoked directly.

Level	Date and Time	Source	Event ID	Task Category
Warning	4/19/2026 10:32:12 AM	PowerShell (Microsof...	4100	Executing Pipeline
Warning	4/19/2026 10:32:12 AM	PowerShell (Microsof...	4100	Executing Pipeline
Warning	4/19/2026 10:32:12 AM	PowerShell (Microsof...	4100	Executing Pipeline
Verbose	4/19/2026 10:32:12 AM	PowerShell (Microsof...	4104	Execute a Remote Co...
Warning	4/19/2026 10:31:32 AM	PowerShell (Microsof...	4104	Execute a Remote Co...
Verbose	4/19/2026 10:31:31 AM	PowerShell (Microsof...	4104	Execute a Remote Co...
Verbose	4/19/2026 10:31:31 AM	PowerShell (Microsof...	4104	Execute a Remote Co...
Verbose	4/19/2026 10:31:29 AM	PowerShell (Microsof...	4104	Execute a Remote Co...
Information	4/19/2026 10:31:29 AM	PowerShell (Microsof...	40962	PowerShell Console S...
Information	4/19/2026 10:31:27 AM	PowerShell (Microsof...	53504	PowerShell Named Pi...
Information	4/19/2026 10:31:25 AM	PowerShell (Microsof...	40961	PowerShell Console S...
Warning	4/19/2026 10:22:10 AM	PowerShell (Microsof...	4104	Execute a Remote Co...
Information	4/19/2026 10:21:13 AM	PowerShell (Microsof...	40962	PowerShell Console S...
Information	4/19/2026 10:21:13 AM	PowerShell (Microsof...	53504	PowerShell Named Pi...
Information	4/19/2026 10:21:13 AM	PowerShell (Microsof...	40961	PowerShell Console S...
Information	4/14/2026 6:00:10 PM	PowerShell (Microsof...	40962	PowerShell Console S...
Information	4/14/2026 6:00:10 PM	PowerShell (Microsof...	53504	PowerShell Named Pi...
Information	4/14/2026 6:00:10 PM	PowerShell (Microsof...	40961	PowerShell Console S...

Figure 1 shows active event logs created by the malware. It executed powershell, abused the ,net framework, injected shell code, encoded the payload, and more.

Kovter Function:

Kovter is often installed through a phishing attack. A malicious email is sent containing the malware; upon opening the file an associated script is run which installs this malware and is then installed on the victim's device. The script begins by writing encrypted code directly into the windows registry before deleting the initially installed executable. This by proxy removes forensic artifacts which can be used to deduce malware function. Obfuscation through removal.

Name	PID	CPU	I/O total ...	Private b...	User name	Description
MoIsoCoreWork...	4396			18.53 MB		MoISO Core Worker Process
WmiPrivSE.exe	7016			2.36 MB		WMI Provider Host
smartscreen.exe	7064			7.58 MB	DESKT...D3play3rWin10	Windows Defender SmartScre...
WmiPrivSE.exe	7116			2.71 MB		WMI Provider Host
mshta.exe	5812			15.67 MB	DESKT...D3play3rWin10	Microsoft (R) HTML Applicati...
powershell.exe	4104			45 MB	DESKT...D3play3rWin10	Windows PowerShell
conhost...	6812			3.33 MB	DESKT...D3play3rWin10	Console Window Host
svchost.exe	880			7.67 MB		Host Process for Windows Ser...
svchost.exe	928			2 MB		Host Process for Windows Ser...
svchost.exe	600			2.19 MB		Host Process for Windows Ser...
svchost.exe	1112			3.14 MB		Host Process for Windows Ser...

Figure 2: A gif showing the malware removing itself from the system.

The code originally written into the registry is base64 and as such cyberchef was used in the process of confirming the decoded base64 text. The malware begins by building a new .net assembly in memory, at run, to remove disk forensic artifacts. In order to get the required elevated permissions for this it looks for the System.dll file in order to append its data too.

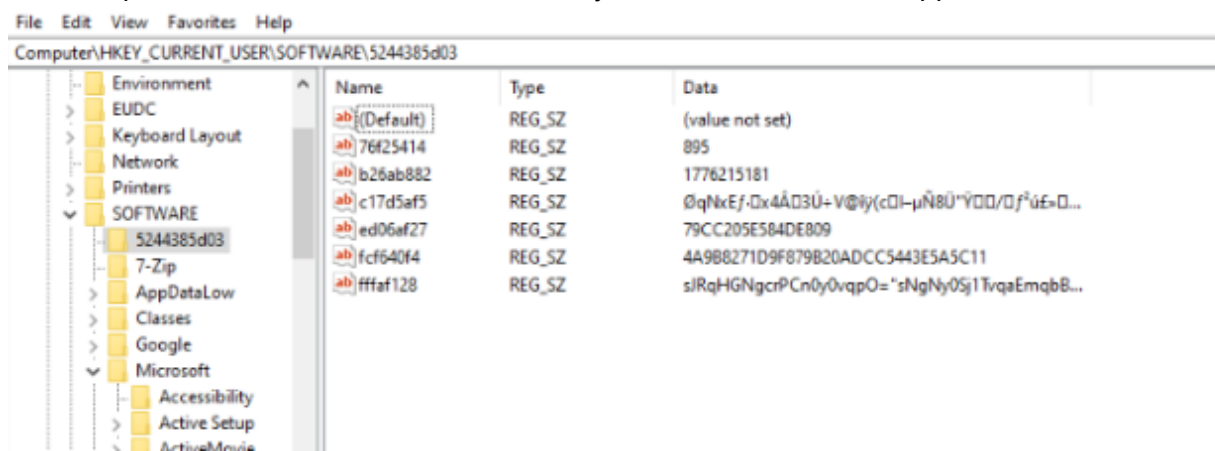


Figure 3: Registry Explorer encoded content created by the malware strain.

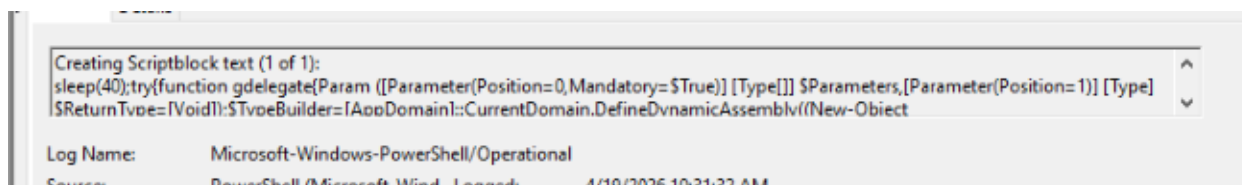


Figure 4: ScriptBlock text showing the creation of .NET assembly

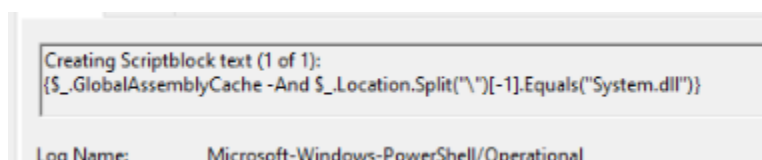


Figure 5: ScriptBlock text showing the search for System.dll

For further explanation this type of malware, as defined in the paper “An emerging threat Fileless malware: a survey and research challenges, would be considered Windows Registry Malware. It installs obfuscated code into a registry key for execution using Powershell and or WMI. Given this malware strain specifically uses Powershell, as seen in Figure 1, this technique can be further defined as Process Hollowing [6]

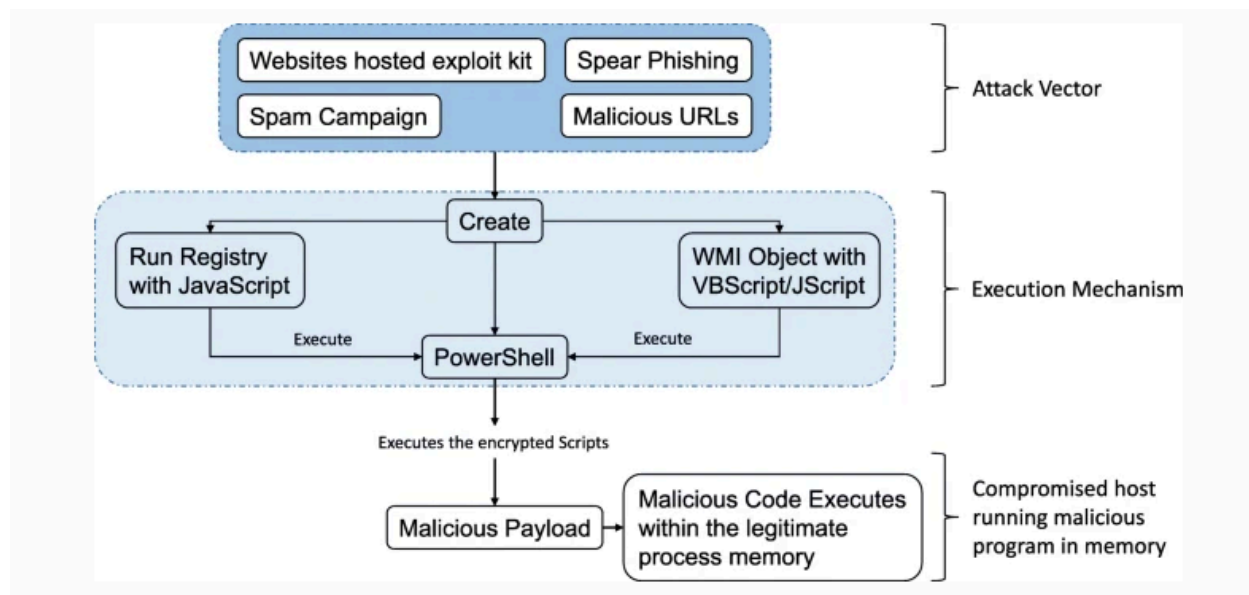


Figure 6: an image depicting the lifecycle of fileless malware. [6]

Indicators of compromise:

Given how the malware works it's important to note that Disk imaging will not be useful as no files can be found. The AV is bypassed as it uses legitimate files to hide itself. File carving will be useless as nothing will be found

Rule Based Detection: [7]

1. If cmd.exe has the parent of winword.exe, excel.exe, powerpnt.exe or outlook.exe and a child process of powershell.exe its likely malware
2. if cmd.exe has the parent of winword.exe, excel.exe, powerpnt.exe or outlook.exe that is cause for suspicion
3. If powershell.exe has the parent process of winword.exe that is cause for suspicion

Tool Based Detection:

1. Process Hacker/Explorer to see suspicious processes, injected threads, and hidden activity in real time
2. Behavioral vs signature detection
3. EDR tools like Microsoft Defender that watch process behavior in real time

General Detection:

1. Look for mshta.exe running unauthorized scripts, or powershell.exe launching unexpectedly
2. Look for suspicious encrypted registry keys
3. Run anti-malware tools such as Windows Defender to identify the malicious registry entries

Conclusions:

This analysis of Kovter shows how serious the threat of fileless malware is. By leveraging legitimate system components like PowerShell, the Windows Registry, and the .NET framework, Kovter is able to execute its payload entirely in memory while leaving minimal disk-based artifacts. This renders traditional forensic techniques such as disk imaging and file carving largely ineffective, meaning a shift towards behavioral and memory-based detection methodologies is needed as these types of malware become increasingly prevalent.